Embedded Storage

FerriSSD® SATA Gen 3 DRAM BGA SSD

Dx Series

Datasheet

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Revision 0.8 Dec 2019

Revision History

Revision	Date	Description
0.1	Aug 31, 2017	Initial release
0.2	Nov 6, 2017	• Updated Key Features (1.2) • Updated signal descriptions of Ball No. F2, G2, and D3
0.3	Dec 13, 2017	• Updated the signal description/type/PU/PD of VCCK, VP, SW_RST#, ENRLS, PGIN#, and PGS# (2.1) (2.2) • Updated the notes of Section (2.2) • Updated the General Operating Conditions table: VCCK and VP typical value (3) • Updated the SMART Data Vendor-specific Attributes Table: Attribute ID A0, A2, C3, C6, C7, C8, FA, and AC (4.3.2)
0.4	Feb 1, 2018	• Unified the wording of ACS-2 command support in the document (1.2) • Updated the signal description and type of the DEVSLP signal (2.2) • Updated the ID Table Information: Word 76, 82 - 87, 119 - 120, and 206 (4.2) • Updated the product coding rule table and top marking figure (6)
0.5	Oct 24, 2018	• Updated the default values in ID Table: Word 59, 69, and 78 (4.2)
0.6	Mar 13, 2019	• Updated the SMART Attributes table: Attribute C2, C8, and FC (4.3.2)
0.7	Aug 13, 2019	• Added optional enhanced security features (1.2) • Updated ID Table Information: Word 48, 59, and table notes (4.2) • Updated FerriSSD Capacity table and added new table: FerriSSD Capacity with TCG feature (4.4) • Added product revision DF and 512GB support in product coding rule table and top marking accordingly (6)
0.8	Dec 12, 2019	• Updated the PU/PD of ball D6 (2.2) • Fixed signal name in the General I/O DC Characteristics table (Table 7) • Added commands in Sanitize Device, Device Configuration Overlay, 48-bit Address and General Feature Set, and added a table note (4.1)

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1. Overview

1.1 Product Description

The FerriSSD® is designed optimally for a wide range of embedded applications requiring faster access speed, small flexible form factor, and reliable SATA storage. By combining industry proven controller technology, NAND flash and passive components into a small single BGA package, FerriSSD® simplifies design efforts, reduces time-to-market while protecting from NAND technology migration concerns.

The new generation FerriSSD Dx series with 3D NAND Flash leveraging Silicon Motion's advanced technologies, including IntelligentScan™, DataRefresh™, high bandwidth LDPC code correction with proprietary RAID engine, and end-to-end data path protection to provide unsurpassed data integrity in a non-volatile storage device. The DRAM FerriSSD stands for the ultimate storage solution for embedded computing devices such as navigation, thin-client, POS, MFP, telecommunications, factory automation, as well as varieties of server applications.

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1.2 Key Features

- Host Interface
  - Industrial Standard SATA Revision 3.1 compliant
  - Industrial Standard ATA/ATAPI-8 and ACS-2 command compliant
  - Supports SATA interface rate of 6Gb/s (backward compatible to 1.5Gb/s and 3Gb/s)
  - Native Command Queuing up to 32 commands
  - SATA Device Sleep (DevSleep)
  - Data Set Management command (TRIM)
  - Supports 28-bit and 48-bit LBA (Logical Block Addressing) mode commands
- Robust Data Protection and Data Reliability
  - Advanced system level protection against unstable power supply
  - SRAM and DRAM ECC protection
  - End-to-end data path protection
  - Hardware LDPC ECC engine with hard-decision and soft-decision decoding
  - RAID engine offers additional level of data protection
  - Internal data shaping optimizes the data endurance
  - StaticDataRefresh and EarlyRetirement technologies ensure data integrity and prevent read disturbance
  - Early weak block retirement feature
  - PowerShield and DataPhoenix technologies support power-down data protection and recovery

- Enhanced Security [1]
  - Real time full drive encryption with Advanced Encryption Standard (AES)
  - Trusted Computing Group (TCG) Opal protocol
  - Hardware SHA 256 and True Random Number Generator (TRNG)
- SSD Status Monitoring
  - Supports Self-Monitoring, Analysis, and Reporting Technology (S.M.A.R.T.) commands
  - Provides SSDLifeGuard application for real time SSD health status monitoring and warning
- Advanced Global Wear Leveling
  - Fully utilizes each cell to even program/erase count across management units/die(s)
  - Maximizes product lifespan with minimal wear leveling and write amplification overhead
- Advanced Security
  - Supports ATA8 security feature set
- Architecture
  - 32-bit RISC CPU
  - High-efficiency 64-bit system bus
  - Automatic sleep and wake-up mechanism to save power
  - Built-in voltage detectors for power failure protection
  - Built-in temperature sensor for SSD temperature detection
  - Supports the JTAG interface and UART (RS-232) interface for on-system debug
- Optional digitally signed firmware with eFuse for added firmware upgrade security
- Easy-to-Use
  - The Plug & Play device only requires format/fdisk prior to use
- Temperature Range
  - Commercial Operating Temperature: 0°C ~ 70°C
  - Extended Operating Temperature: -40°C ~ +85°C
  - Non-Operating and Storage Temperature: -55°C ~ +85°C
- Package
  - Small Form Factor: 16mm x 20mm
  - Green Package
  - RoHS Compliant

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1 Optional features are activated by specific firmware version.

1.3 Functional Description

Data Transfer

The flash controller enables multi-way interleaving for multi-bank flash connection to obtain optimal performance. The FerriSSD uses a superior DMA technology to transfer data between the host and the NAND flash interface. The DMA technology transfers data at a very high rate in both directions (read and write) and in doing so, effectively decreases the loading of micro processor.

ECC Correction

The LDPC ECC engine executes parity generation and error detection/correction features, and enhances decoding throughput and data reliability. With LDPC of correction capability 1e-2 RBER, the hard and soft decoding mechanism provides powerful error correction. Hence the FerriSSD can enhance the endurance and retention of SSD.

RAID Protection

In case of uncorrectable errors occurring within a superblock (a pre-defined area which consists of a particular set of blocks across physical NAND units), the RAID engine recovers the uncorrectable error chunk by using a certain storage space of parity bits. Incorporated with LDPC, the RAID ensures a comprehensive level of data integrity while providing a broad range of RAID overhead protection.

SMART and Data Security

The FerriSSD supports SMART commands that allow users to read spare and bad block information. The users can thus evaluate drive health at run time and receive an early warning before the flash drive lifespan ends. In addition, the users can use security commands to lock and unlock the drive by password or a hardware switch and protect sensitive data stored in the SSD.

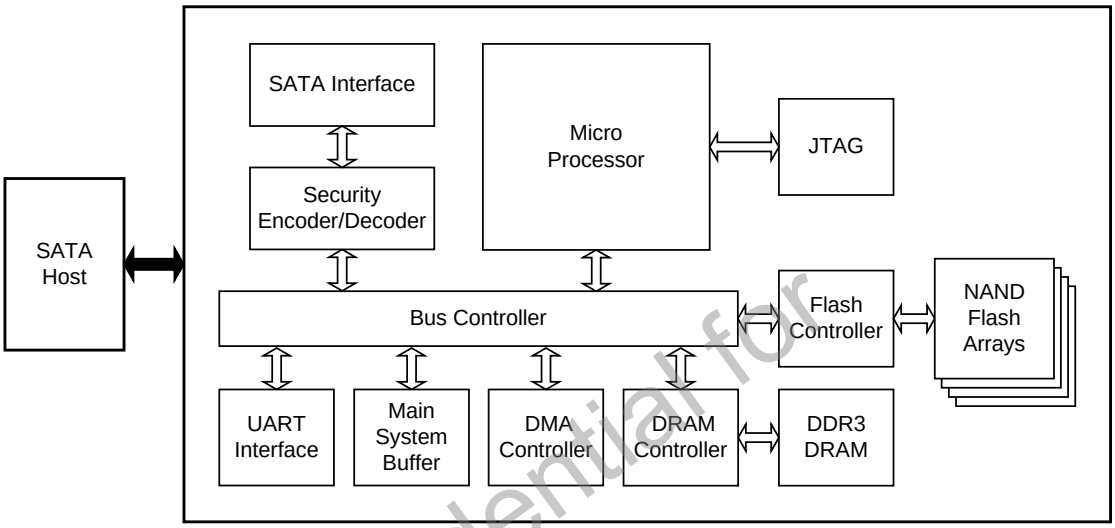
Power Management

The FerriSSD supports SATA DEVSLP to greatly save power consumption. In addition, the embedded PLL circuits run the system clock and the frequency can be programmed to fit different NAND flash timings. With the power-on reset (POR) circuit, the SSD provides a reset signal for the core logic after power on.

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1.4 Block Diagram

Figure 1: FerriSSD Block Diagram

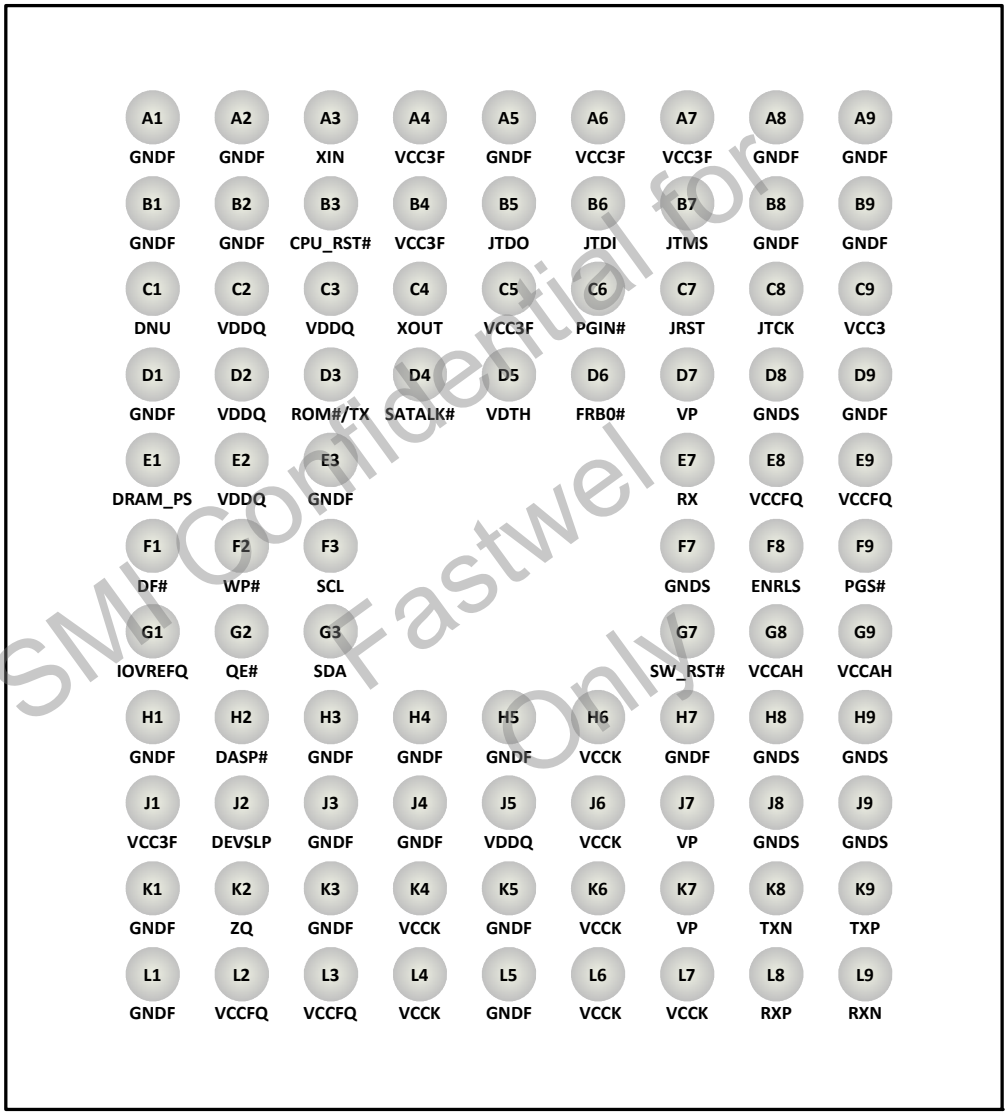


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2. Signal Descriptions

2.1 Ball Assignments

Figure 2: 90-Ball Assignments (Top View - Balls down)



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2.2 Signal Description Table

Table 1: SATA Interface Signals

Signal	Type	PU/PD	Ball No.	Description
RXP	I	-	L8	Positive input of receiver differential signal
RXN	I	-	L9	Negative input of receiver differential signal
TXP	O	-	K9	Positive output of transmitter differential signal
TXN	O	-	K8	Negative output of transmitter differential signal
XIN	I	-	A3	50MHz external crystal input
XOUT	O	-	C4	50MHz external crystal output

Table 2: Power/Ground Signals

Signal	Type	PU/PD	Ball No.	Description
VCC3	PWR	-	C9	Supply voltage 3.3V for general I/O
VCC3F	PWR	-	A4, A6, A7, B4, C5, J1	Supply voltage 3.3V for flash core
VCCFQ	PWR	-	E8, E9, L2, L3	Supply voltage 1.8V for flash I/O
VCCK	PWR	-	H6, J6, K4, K6, L4, L6, L7	Supply voltage 1.1V for core logic
VCCAH	PWR	-	G8, G9	Analog supply voltage 3.3V
VP	PWR	-	D7, J7, K7	Analog supply voltage 1.1V for Tx/Rx/AIP
VDTH	PWR	-	D5	Voltage detector input for host power When host power is 5V, this signal connects to the host power supply. When host power is 3.3V, keep this signal floating.
VDDQ	PWR	-	C2, C3, D2, E2, J5	Supply voltage 1.5V for DRAM
IOVREFQ	PWR	-	G1	DRAM reference voltage (VDDQ x 0.5)
GNDF	GND	-	A1, A2, A5, A8, A9, B1, B2, B8, B9, D1, D9, E3, H1, H3, H4, H5, H7, J3, J4, K1, K3, K5, L1, L5	Digital ground connection.
GNDS	GND	-	D8, F7, H8, H9, J8, J9	Analog ground connection

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Table 3: Functional Signals

Signal	Type	PU/PD	Ball No.	Description
FRB0#	I	-	D6	Flash ready/busy. This pin only needs an external 2kΩ pull-up resistor.
CPU_RST#	I	PU	B3	Hardware reset. To properly configure the corresponding relation between power on sequence and hardware reset, it is recommended to follow the instructions of the Ferri DRAM Power On_off Sequence application note.
DASP#	I/O	PU[1]	H2	Device activity (low active). The DASP# can be configured to be input or output signal by the firmware (output by default). When this signal is driven low by the device, it indicates that commands are issued to the device (into busy state). For example, it can be used for LED indicator to indicate the busy status of the device. Keep this signal floating in case of not in use.
DF#	I/O	PU[1]	F1	Hardware data flush (low active). The DF# can be configured to be input or output signal by the firmware (input by default). When this signal is driven low, the device flushes data from its cache buffer to the flash memory. To support this optional function or not is depended by customers. Keep this signal floating in case of not in use.
SATALK#	I/O	PU[1]	D4	SATA link status (low active). The SATALK# can be configured to be input or output signal by the firmware (output by default). When the device drives this signal low, it indicates that the SATA link is active. For example, it can be used for LED indicator to indicate the SATA link status. Keep this signal floating in case of not in use.
DEVSLP	I	-	J2	SATA device sleep mode (DevSleep). An external 33kΩ pull-up resistor must be added to the DEVSLP according the requirement of Serial ATA specification. When the host enables the DevSleep mode, this pin will be driven high and the device enters the DevSleep power state. To support this optional function or not is depended by customers. Keep this signal floating in case of not in use.

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Signal	Type	PU/PD	Ball No.	Description
DRAM_PS	I/O	PU[1]	E1	Enable power supply (MOS switch) for DRAM. The DRAM_PS I/O is controlled by firmware (output by default). If the host supports the device sleep feature, the firmware will configure this signal as output. Firmware drives this signal low when the device enters the DevSleep interface power state. To support this optional function or not is depended by customers. Keep this signal floating in case of not in use.
ZQ	I	-	K2	External reference for output drive calibration. This pin only needs an external 240Ω pull-down resistor.
SW_RST#	I/O	PU[1]	G7	Software reset (low active). The SW_RST# can be configured to be input or output signal by the firmware (input by default). When this signal is driven low, the device performs software reset for CPU core/SATA/NAND Flash/DRAM components.
ENRLS	I/O	PD[1]	F8	Reserved for DF circuit
PGIN#	I/O	PU[1]	C6	Reserved for DF circuit
PGS#	I/O	PU[1]	F9	Reserved for DF circuit
WP#	I/O	PU[1]	F2	Reserved for special purpose. Keep this signal floating for normal operation.
QE#	I/O	PU[1]	G2	Reserved for special purpose. Keep this signal floating for normal operation.
DNU	/	-	C1	Do not use. DNU pins must be left disconnected

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Table 4: Test/Debug Signals

Signal	Type	PU/PD	Ball No.	Description
ROM# /TX	I/O	PU[1]	D3	Reserved for special purpose. Keep this signal floating for normal operation.
RX	I/O	PU[1]	E7	For UART RX input (for debug only) For debug purpose, it is recommended to connect this pin to test pad. This pin can be left floating if it is not used.
JRST	I	PU[1]	C7	JTAG test reset (for debug only) For debug purpose, it is recommended to connect this pin to test pad. This pin can be left floating if it is not used.
JTMS	I	PU[1]	B7	JTAG test mode select (for debug only) For debug purpose, it is recommended to connect this pin to test pad. This pin can be left floating if it is not used.
JTDI	I	PU[1]	B6	JTAG test data input (for debug only) For debug purpose, it is recommended to connect this pin to test pad. This pin can be left floating if it is not used.
JTDO	O	-	B5	JTAG test data output (for debug only) For debug purpose, it is recommended to connect this pin to test pad. This pin can be left floating if it is not used.
JTCK	I	PU[1]	C8	JTAG test clock (for debug only) For debug purpose, it is recommended to connect this pin to test pad. This pin can be left floating if it is not used.
SCL	I/O	PU	F3	Serial bus clock (for debug only) For debug purpose, it is recommended to connect this pin to test pad. This pin can be left floating if it is not used.
SDA	I/O	PU	G3	Serial bus data (for debug only) For debug purpose, it is recommended to connect this pin to test pad. This pin can be left floating if it is not used.

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Notes:

1. For the pull-up/pull-down resistance values, refer to Table 7.
   PU[1]: Internal pulled-up by firmware when the signal is configured as input.
   PU: Internal fixed pull-up.
   PD[1]: Internal pulled-down by firmware when the signal is configured as input.
   PD: Internal fixed pull-down.
2. Low active signals have a "#" suffix.
3. For more information on signals, please contact project manager or local support personnel.

3. Electrical Characteristics

Table 5: Absolute Maximum Ratings

Parameter	Symbol	Min	Typ	Max	Unit
General I/O Supply Voltage	VCC3	-0.3		4.6	V
Digital Power Supply	VCCK	-0.3		1.4	V
Analog Power Supply	VCCAH	-0.3		4.6	V
Flash Core Power Supply	VCC3F	-0.6		4.6	V
Flash I/O Power Supply	VCCFQ	-0.2		2.4	V
DRAM Power Supply	VDDQ	-0.5		1.98	V
SATA PHY Power Supply	VP	-0.5		1.21	V

Table 6: General Operating Conditions

Parameter	Symbol	Min	Typ	Max	Unit
General I/O Supply Voltage	VCC3	2.7	3.3	3.6	V
Digital Power Supply	VCCK	0.99	1.1	1.21	V
Analog Power Supply	VCCAH	3.06	3.3	3.63	V
Flash Core Power Supply	VCC3F	2.85	3.3	3.6	V
Flash I/O Power Supply	VCCFQ	1.7	1.8	1.95	V
DRAM Power Supply	VDDQ	1.43	1.5	1.57	V
SATA PHY Power Supply	VP	1.023	1.1	1.21	V

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Table 7: General I/O DC Characteristics

Parameter	Symbol	Condition	Min	Typ	Max	Unit
Supply Voltage	VCC3		2.7	3.3	3.6	V
High Level Output Voltage	V_{OH}		0.67 x VCC3			V
Low Level Output Voltage	V_{OL}				0.4	V
High Level Input Voltage	V_{IH1}		0.8 x VCC3		VCC3 + 0.3	V
Low Level Input Voltage	V_{IL1}		GNDF - 0.3		0.2 x VCC3	V
High Level Input Voltage	V_{IH2}		0.55 x VCC3		VCC3 + 0.3	V
Low Level Input Voltage	V_{IL2}		GNDF - 0.3		0.48 x VCC3	V
Pull-up Resistance	R_{PU}		28.6	40.8	69.4	kΩ
Pull-down Resistance	R_{PD}		27.1	38.7	69.4	kΩ

Notes:

1. V_{IH1} and V_{IL1} are applicable to general I/O.
2. V_{IH2} and V_{IL2} are applicable to CPU reset.

Table 8: Power-on Reset with Voltage Detection

Parameter		Min	Typ	Max	Unit
Voltage Detection Threshold	Rising	0.79	0.93	0.995	V
	Falling	0.706	0.76	0.813	V
Voltage Detection Delay Time	Rising	2.12	2.62	3.27	µs
	Falling	4.06	5.32	6.96	µs

Table 9: 2.7V Voltage Detector

Parameter		Min	Typ	Max	Unit
Voltage Detection Threshold	Rising	2.7	2.8	2.9	V
	Falling	2.6	2.7	2.8	V
Voltage Detection Delay Time	Rising			4.5	µs
	Falling			1.5	µs

Table 10: 4.0V Voltage Detector

Parameter		Min	Typ	Max	Unit
Voltage Detection Threshold	Rising	4.00		4.28	V
	Falling	3.86		4.14	V
Voltage Detection Delay Time	Rising	1.5		4.5	µs
	Falling	0.5		1.5	µs

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Table 11: PLL

Parameter	Symbol	Min	Typ	Max	Unit
Output Clock	PLL_OUT/ PHASE1~4	12.5		400	MHz
Duty Cycle		45		55	%
Jitter				10%	UI[1]
PLL Lock Time			101.2		µs

Note[1]: UI = output frequency.

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4. Software Interface

The following tables provide the contents of FerriSSD current specifications. The contents may vary in the latest firmware version.

4.1 FerriSSD Command Set

Table 12: Command Set

Command	Code	Protocol
General Feature Set		
EXECUTE DEVICE DIAGNOSTIC	90h	Execute device diagnostic
FLUSH CACHE	E7h	Non-data
IDENTIFY DEVICE	ECh	PIO data-in
INITIALIZE DRIVE PARAMETERS	91h	Non-data
READ DMA	C8h	DMA
READ LOG EXT	2Fh	PIO data-in
READ LOG DMA EXT	47h	DMA
READ MULTIPLE	C4h	PIO data-in
READ SECTOR(S)	20h	PIO data-in
READ VERIFY SECTOR(S)	40h or 41h	Non-data
WRITE UNCORRECTABLE EXT	45h	Non-data
SET FEATURE	EFh	Non-data
SET MULTIPLE MODE	C6h	Non-data
WRITE DMA	CAh	DMA
WRITE LOG EXT	3Fh	PIO data-out
WRITE LOG DMA EXT	57h	DMA
WRITE MULTIPLE	C5h	PIO data-out
WRITE SECTOR(S)	30h	PIO data-out
NOP	00h	Non-data
READ BUFFER	E4h	PIO data-in
READ BUFFER DMA	E9h	DMA
WRITE BUFFER	E8h	PIO data-out
WRITE BUFFER DMA	EBh	DMA
DOWNLOAD MICROCODE	92h	PIO data-out/Non-Data
DOWNLOAD MICROCODE DMA	93h	DMA/Non-Data
Power Management Feature Set		
CHECK POWER MODE	E5h or 98h	Non-data
IDLE	E3h or 97h	Non-data
IDLE IMMEDIATE	E1h or 95h	Non-data

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Command	Code	Protocol
SLEEP	E6h or 99h	Non-data
STANDBY	E2h or 96h	Non-data
STANDBY IMMEDIATE	E0h or 94h	Non-data
Security Feature Set		
SECURITY SET PASSWORD	F1h	PIO data-out
SECURITY UNLOCK	F2h	PIO data-out
SECURITY ERASE PREPARE	F3h	Non-data
SECURITY ERASE UNIT	F4h	PIO data-out
SECURITY FREEZE LOCK	F5h	Non-data
SECURITY DISABLE PASSWORD	F6h	PIO data-out
SMART Feature Set		
SMART DISABLE OPERATIONS	B0h	Non-data
SMART ENABLE/DISABLE AUTOSAVE	B0h	Non-data
SMART ENABLE OPERATIONS	B0h	Non-data
SMART EXECUTE OFF-LINE IMMEDIATE	B0h	Non-data
SMART READ LOG	B0h	PIO data-in
SMART READ DATA	B0h	PIO data-in
SMART READ THRESHOLD	B0h	PIO data-in
SMART RETURN STATUS	B0h	Non-data
SMART SAVE ATTRIBUTE VALUES	B0h	Non-data
SMART WRITE LOG	B0h	PIO data-out
Host Protected Area Feature Set		
READ NATIVE MAX ADDRESS	F8h	Non-data
SET MAX ADDRESS	F9h	Non-data
SET MAX SET PASSWORD	F9h	PIO data-out
SET MAX LOCK	F9h	Non-data
SET MAX FREEZE LOCK	F9h	Non-data
SET MAX UNLOCK	F9h	PIO data-out
48-bit Address Feature Set		
FLUSH CACHE EXT	EAh	Non-data
READ SECTOR(S) EXT	24h	PIO data-in
READ DMA EXT	25h	DMA
READ MULTIPLE EXT	29h	PIO data-in
READ NATIVE MAX ADDRESS EXT	27h	Non-data
READ VERIFY SECTOR(S) EXT	42h	Non-data
SET MAX ADDRESS EXT	37h	Non-data
WRITE DMA EXT	35h	DMA
WRITE DMA FUA EXT	3Dh	DMA

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Command	Code	Protocol
WRITE MULTIPLE EXT	39h	PIO data-out
WRITE MULTIPLE FUA EXT	CEh	PIO data-out
WRITE SECTOR(S) EXT	34h	PIO data-out
NCQ Feature Set		
READ FPDMA QUEUED	60h	DMA Queued
WRITE FPDMA QUEUED	61h	DMA Queued
Sanitize Device Feature Set		
BLOCK ERASE EXT	B4h	Non-data
CRYPTO SCRAMBLE EXT [1]	B4h	Non-data
SANITIZE FREEZE LOCK EXT	B4h	Non-data
SANITIZE STATUS EXT	B4h	Non-data
Device Configuration Overlay Feature Set [1]		
DEVICE CONFIGURATION FREEZE LOCK	B1h	Non-data
DEVICE CONFIGURATION IDENTIFY	B1h	PIO data-in
DEVICE CONFIGURATION IDENTIFY DMA	B1h	DMA
DEVICE CONFIGURATION RESTORE	B1h	Non-data
DEVICE CONFIGURATION SET	B1h	PIO data-out
DEVICE CONFIGURATION SET DMA	B1h	DMA
Trusted Computing Feature Set [1]		
TRUSTED NON-DATA	5Bh	Non-Data
TRUSTED SEND	5Eh	PIO data-out
TRUSTED SEND DMA	5Fh	DMA
TRUSTED RECEIVE	5Ch	PIO data-in
TRUSTED RECEIVE DMA	5Dh	DMA
Others		
DATA SET MANAGEMENT	06h	DMA
SEEK	70h	Non-data

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Note [1]: The command will be activated by MP option.

4.2 Identify Device

The Identify Device command enables the host to receive parameter information from the FerriSSD. This command has the same protocol as the Read Sector(s) command. The parameter words in the buffer have the arrangement and meanings defined in the following table.

Table 13: ID Table Information

Word	F / V	Default Value	Description
0	F	0040h	General configuration
1	X	XXXXh	Default number of cylinders
2	V	C837h	Specific configuration
3	X	00XXh	Default number of heads
4	X	0000h	Obsolete
5	X	0000h	Retired
6	F	XXXXh	Default number of sectors per track
7 - 8	X	0000h	Retired
9	X	0000h	Obsolete
10 - 19	F	XXXXh	Serial number in ASCII (Right justified)
20 - 21	X	0000h	Retired
22	X	0000h	Obsolete
23 - 26	F	XXXXh	Firmware revision in ASCII Big Endian Byte Order in Word
27 - 46	F	XXXXh	Model number in ASCII (Left justified) Big Endian Byte Order in Word
47	F	8002h	Maximum number of sectors on Read/Write Multiple command
48	F	4000h	Reserved
		4001h	Trusted Computing feature set is supported
49	F	2F00h	Capabilities
50	F	4000h	Capabilities
51	X	0000h	Obsolete
52	X	0000h	Obsolete
53	F	0007h	Field validity
54	X	XXXXh	Current numbers of cylinders
55	X	XXXXh	Current numbers of heads
56	X	XXXXh	Current sectors per track
57 - 58	X	XXXXh	Current capacity in sectors (LBAs) (Word 57 = LSW , Word 58 = MSW)

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Word	F / V	Default Value	Description
59	F/V	91XXh	Multiple sector setting is valid ATA Sanitize Device features with Block Erase method is supported. Cryptographic Scramble method is not supported.
		B1XXh	Multiple sector setting is valid ATA Sanitize Device features with Block Erase method is supported. Cryptographic Scramble method is supported.
60 - 61	F	XXXXh	Total number of user addressable logical sectors for 28-bit commands (DWord)
62	X	0000h	Reserved
63	F/V	0407h	Multiword DMA transfer Supports MDMA mode 0, 1 and 2
64	F	0003h	Advanced PIO modes supported
65	F	0078h	Minimum Multiword DMA transfer cycle time per word
66	F	0078h	Recommended Multiword DMA transfer cycle time
67	F	0078h	Minimum PIO transfer cycle time without flow control
68	F	0078h	Minimum PIO transfer cycle time with IORDY flow control
69	F	4D20h	Additional supported
70 - 74	F	0000h	Reserved
75	F	001Fh	Queue depth
76	F	830Eh	Serial ATA capabilities
77	F/V	0084h	Serial ATA additional capability
78	F	004Ch	Serial ATA features supported
79	V	0040h	Serial ATA features enabled
80	F	03F0h	Major version number (ACS-2)
81	F	0000h	Minor version number
82	F	746Bh	Command sets supported
83	F	7501h	Command sets supported
84	F	4063h	Command sets supported
85 - 87	V	XXXXh	Command set/feature enabled
88	V	007Fh	Ultra DMA mode supported and selected
89	F	0001h	Time required for a Normal Erase mode Security Erase Unit command
90	F	0001h	Time required for an Enhanced Erase mode Security Erase Unit command
91	V	0000h	Current advanced power management value
92	V	FFFEh	Master password identifier
93 - 99	V	0000h	Reserved
100 - 103	V	XXXXh	Maximum user LBA for 48-bit address feature set
104	V	0000h	Reserved
105	F	0008h	Maximum number of 512-byte blocks per Data Set Management command
106	V	4000h	Physical sector size/logical sector size

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Word	F / V	Default Value	Description
107 - 116	V	0000h	Reserved
117 - 118	F	00000100h	Logical sector size
119	F	401Ch	Command sets supported
120	V	401Ch	Command set/feature enabled
121 - 127	V	0000h	Reserved
128	V	0021h	Security status
129 - 159	X	XXXXh	Vendor specific
160 - 167	V	0000h	Reserved
168	F	0000h	Nominal form factor not reported
169	F	0001h	Data Set Management supported
170 - 205	V	XXXXh	Reserved
206	F	0035h	SCT Command Transport
207 - 208	V	XXXXh	Reserved
209	F	4000h	Alignment of logical sectors within a physical sector
210 - 216	V	XXXXh	Reserved
217	F	0001h	Non-rotating media (SSD)
218 - 221	X	0000h	Reserved
222	F	107Fh	Transport major revision (SATA Rev 3.1)
223 - 233	X	0000h	Reserved
234	F	0001h	Minimum number of 512-byte data blocks per Download Microcode operation
235	F	0200h	Maximum number of 512-byte data blocks per Download Microcode operation
236 - 254	X	0000h	Reserved
255	X	XXXXh	Integrity word

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Notes:

1. F = Content (byte) is fixed and does not change.
2. V = Content (byte) is variable and may change depending on the state of the device or the commands executed by the device.
3. X = Content (byte) is vendor specific and may be fixed or variable.
4. TCG support is activated by MP option.
5. Cryptographic Scramble method support is activated by MP option.

4.3 SMART Feature Set

The FerriSSD supports the SMART command set and defines some vendor-specific data to report spare/bad block numbers in each memory management unit.

Table 14: SMART Feature Register Values

Value	Command	Value	Command
D0h	Read Data	D5h	Read Log
D1h	Read Attribute Threshold	D6h	Write Log
D2h	Enable/Disable Autosave	D8h	Enable SMART Operations
D3h	Save Attribute Values	D9h	Disable SMART Operations
D4h	Execute OFF-LINE Immediate	DAh	Return Status

If the reserved size is below the threshold, the status can be read from the Cylinder Register using the Return Status command (DAh).

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4.3.1 SMART Data Structure

The following 512 bytes make up the device SMART data structure. Users can obtain the data using the "Read Data" command (D0h).

Table 15: SMART Data Structure

Byte	F / V	Description
0 - 1	X	Revision code
2 - 361	X	Vendor specific (see 4.3.2)
362	V	Off-line data collection status
363	X	Self-test execution status byte
364 - 365	V	Total time in seconds to complete off-line data collection activity
366	X	Vendor specific
367	F	Off-line data collection capability
368 - 369	F	SMART capability
370	F	Error logging capability • 7-1 Reserved • 0 1 = Device error logging supported
371	X	Vendor specific
372	F	Short self-test routine recommended polling time (in minutes)
373	F	Extended self-test routine recommended polling time (in minutes)
374	F	Conveyance self-test routine recommended polling time (in minutes)
375 - 376	F	Extended self-test routine recommended polling time (in minutes)
377 - 385	R	Reserved
386 - 505	R	Vendor specific (see 4.3.2)
506 - 510	X	Reserved
511	V	Data structure checksum

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Notes:

1. F = content (byte) is fixed and does not change.
2. V = content (byte) is variable and may change depending on the state of the device or the commands executed by the device.
3. X = content (byte) is vendor specific and may be fixed or variable.
4. R = content (byte) is reserved and shall be zero.

4.3.2 SMART Attributes

The following table defines the vendor specific data in byte 2 to 361 and byte 386 to 505 of the 512-byte SMART data.

Table 16: SMART Data Vendor-specific Attributes

Attribute ID (hex)	Raw Attribute Value						Attribute Name
01	LSB	00	00	00	00	00	Read Error Rate
05	LSB	MSB	00	00	00	00	Reallocated Logic Block Count
06	LSB	MSB	00	00	00	00	Reallocated Physical Block Count
09	LSB			MSB	00	00	Power-on Hours
0C	LSB			MSB	00	00	Power Cycle Count
A0	LSB			MSB	00	00	On-line Uncorrectable Sector Count (each unit = 4KB)
A1	LSB	MSB	00	00	00	00	Number of Valid Spare Block
A2	LSB	MSB	00	00	00	00	Number Of Running Spare Block
A3	LSB	MSB	00	00	00	00	Number of Initial Invalid Block
A4	LSB			MSB	00	00	Total Erase Count
A5	LSB			MSB	00	00	Maximum Erase Count
A6	LSB			MSB	00	00	Minimum Erase Count
A7	LSB			MSB	00	00	Average Erase Count
A8	LSB			MSB	00	00	Maximum Allowed Erase Count
A9	LSB	00	00	00	00	00	Remaining Life Percentage Based on P/E
AA	LSB	00	00	00	00	00	Remaining Life Percentage Based on Spare count
C0	LSB			MSB	00	00	Sudden Power-off Count
C1	LSB	MSB	00	00	00	00	Off-line Refresh Block Count
C2	LSB						Current Temperature
		LSB					Current Temperature of NAND [1]
			LSB				Minimum Temperature In Lifetime
				LSB			Minimum Temperature In Lifetime of NAND [1]
					LSB		Maximum Temperature In Lifetime
						LSB	Maximum Temperature In Lifetime of NAND [1]
C3	LSB			MSB	00	00	Hardware Enter Hard Decode Count
C4	LSB	MSB	00	00	00	00	On-line Refresh Block Count
C5	LSB	MSB	00	00	00	00	On-line Retire Block Count
C6	LSB			MSB	00	00	Off-line Uncorrectable Sector Count (each unit = 1KB)
C7	LSB			MSB	00	00	SATA Fetal Error Count

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Attribute ID (hex)	Raw Attribute Value						Attribute Name
C8	LSB						System Block Status
		LSB					Bit 3:0 Current Thermal Throttling Mode Bit 7:4 Maximum Thermal Throttling Mode [2]
C9	LSB			MSB	00	00	Hardware Enter Soft Decode Count
CA	LSB			MSB	00	00	Hardware Enter RAID Count
CB	LSB	MSB	00	00	00	00	Off-line Retire Block Count
F1	LSB					MSB	Total LBAs Written (each write unit = 32MB)
F2	LSB					MSB	Total LBAs Read (each read unit = 32MB)
F3	LSB			MSB	00	00	User Capacity
F4	LSB			MSB	00	00	Wear Leveling Count
F5	LSB			MSB	00	00	Garbage Clean Count
F6	LSB			MSB	00	00	Program Fail Count
F7	LSB			MSB	00	00	Erase Fail Count
F8	LSB			MSB	00	00	System Recovery Count
F9	LSB			MSB	00	00	Total Flash Programmed Count (each unit = 32MB)
FA	LSB	MSB					SATA Gen3 Link Count
			LSB	MSB			SATA Gen2 Link Count
					LSB		SATA Gen1 Link Count
						LSB	Overflow Count
FB	LSB			MSB	00	00	Reserved
FC	LSB						SRAM Correctable Count
		LSB					SRAM Uncorrectable Count
			LSB				DRAM Correctable Count
				LSB			DRAM Uncorrectable Count
					LSB	MSB	E2E Error Count

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Notes:

[1]: The attribute will be activated by Firmware version XXS2Ex.

[2]: The attribute will be activated by Firmware version XXS2Ex. The attribute record is only for a single power-on, not across power cycle.

4.4 Capacity

The following tables present the default storage capacity and settings for cylinders, heads, and sectors. Users can change these settings using SMI mass production software.

Table 17: FerriSSD Capacity

Raw Capacity	Cylinders	Heads	Sectors	Total Sectors in LBA Mode
4GB	7,671	16	63	7,732,368
8GB	15,343	16	63	15,465,744
16GB	16,383	16	63	30,932,992
32GB	16,383	16	63	61,865,984
64GB	16,383	16	63	123,731,968
128GB	16,383	16	63	247,463,936
256GB	16,383	16	63	468,862,128
512GB	16,383	16	63	937,703,088

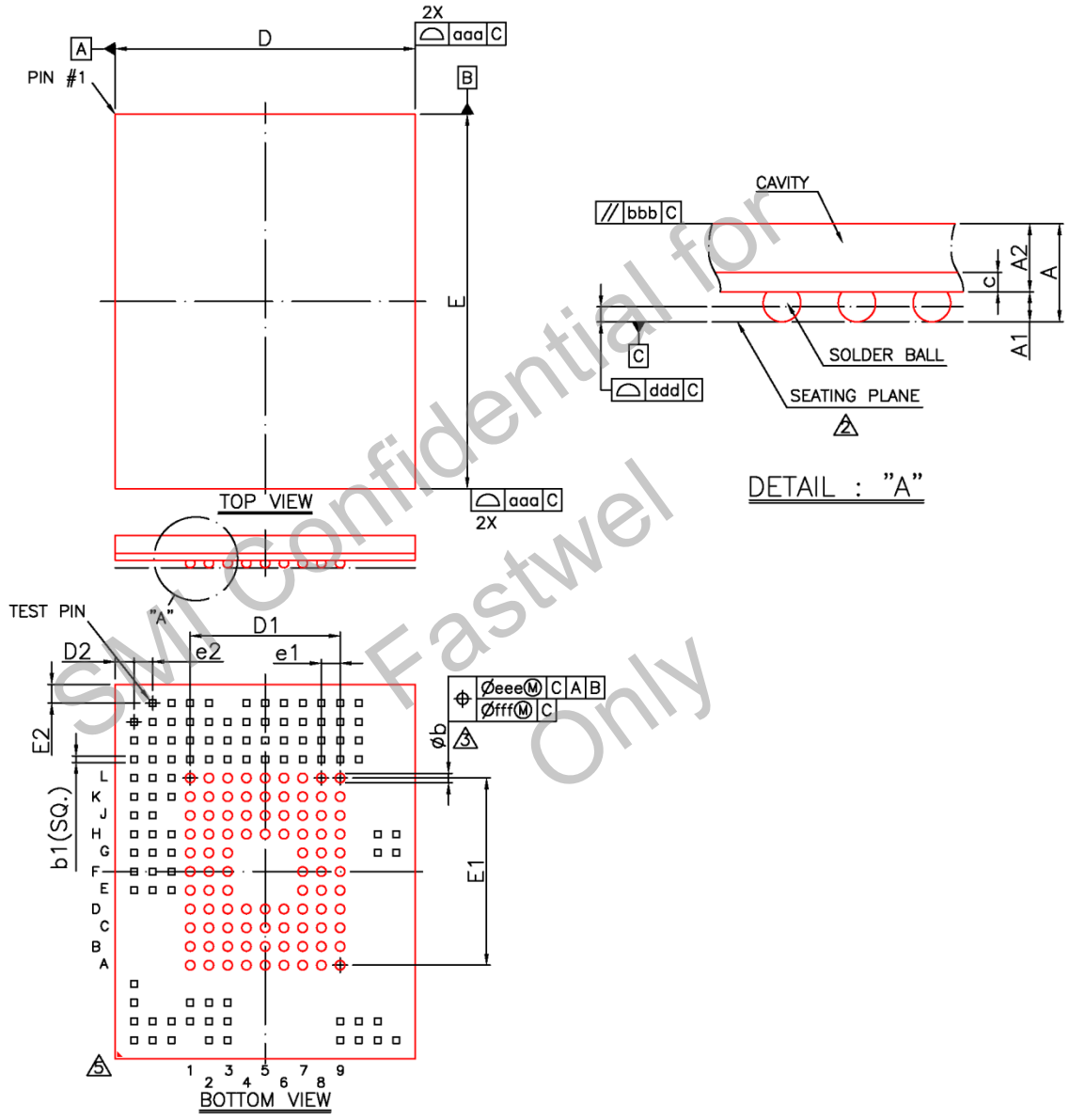
Table 18: FerriSSD Capacity with TCG Feature

Raw Capacity	Cylinders	Heads	Sectors	Total Sectors in LBA Mode
16GB	16,383	16	63	30,367,744
32GB	16,383	16	63	61,300,736
64GB	16,383	16	63	123,166,720
128GB	16,383	16	63	246,898,688
256GB	16,383	16	63	468,296,880
512GB	16,383	16	63	937,137,840

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5. Mechanical Dimensions

5.1 FerriSSD Dx Series Package Outline (16 x 20 mm)



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Symbol	MIN	NOM	MAX	Symbol	MIN	NOM	MAX
A	1.67	1.74	1.81	e1/e2	---	1.00	---
A1	0.35	0.40	0.45	b	0.45	0.50	0.55
A2	1.29	1.34	1.39	b1	---	0.35	---
c	0.32	0.36	0.40	aaa	0.15		
D	15.90	16.00	16.10	bbb	0.20		
E	19.90	20.00	20.10	ddd	0.20		
D1	---	8.00	---	eee	0.15		
E1	---	10.00	---	fff	0.08		
D2	---	1.00	---	MD/ME	9/11		
E2	---	1.00	---	---			

Notes:
1. Controlling dimension: millimeter.
2. Primary datum C and seating plane are defined by the spherical crowns of the solder balls.
3. Dimension b is measured at the maximum solder ball diameter, parallel to primary datum C.
4. Special characteristics C class: bbb, ddd.
5. The pattern of Pin 1 fiducial is for reference only.
6. Ball pad/Ball pad opening size: 500/400 um
7. This package is 90 balls + 97 test pins.

6. Ordering Information

6.1 Product Coding Rule

Table 19: Product Code Definitions

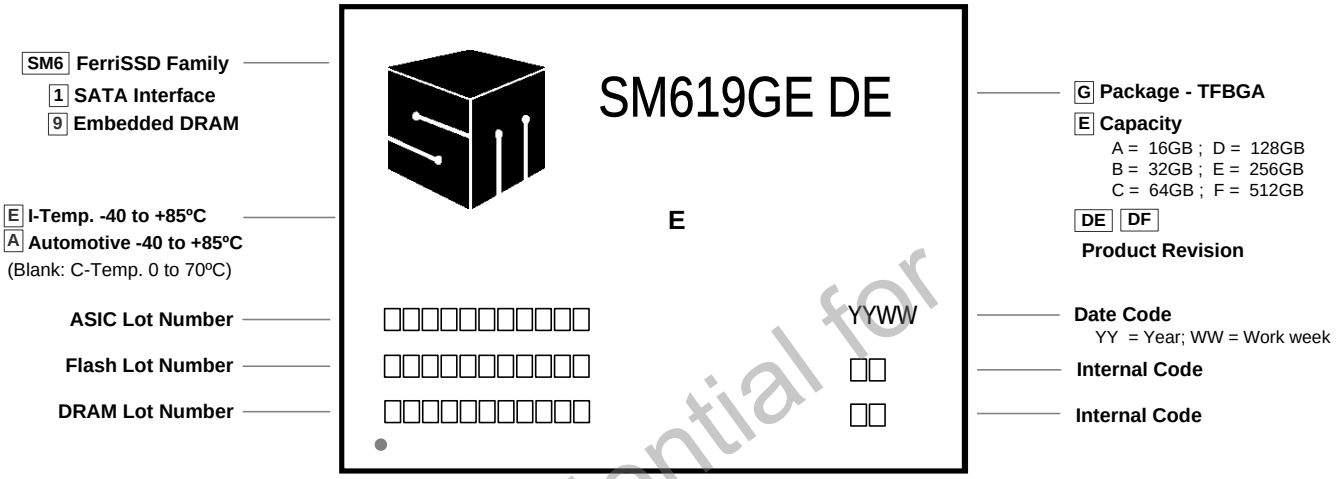
Example: SM 6 1 9 G E E □ DE □	
SM	Silicon Motion
6	Ferri Family
1	SATA Interface
9	9 = Embedded DRAM
G	Package: MCM TFBGA
E	Operating Temperature • X = 0°C ~ 70°C (C-temp) • E = -40°C ~ +85°C (I-temp) • A = -40°C ~ +85°C °C (Automotive AEC-Q100 Grade 3)
E	Capacity • A = 16GB • B = 32GB • C = 64GB • D = 128GB • E = 256GB • F = 512GB
□	Blank (Reserved)
DE/DF	Product Revision
□	Blank (‘T’ or ‘S’ for specified NAND vendor)

Note: See FerriSSD Product Selection Guide for specific ordering numbers.

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6.2 Top Marking

Figure 3: FerriSSD Top Marking (Example)



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